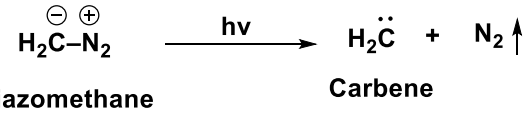
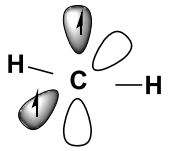
● Cyclopropanation: Reactions of carbenes

- Carbenes are the compounds containing a neutral divalent carbon with 6 electrons

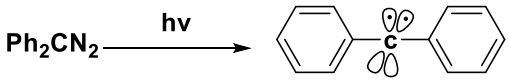


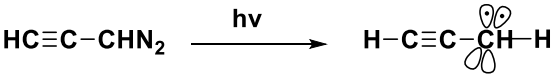
● Structure and Bonding



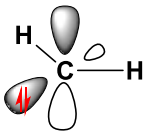
Triplet **M = 2S + 1**
= 2(1/2 + 1/2) + 1
= 3

- **H C H bond angle 130 - 150°**
- **act as biradical**
- **usually generated photochemically**
- **examples -$\ddot{C}H_2$, $\ddot{C}HPh$, $\ddot{C}HR$**



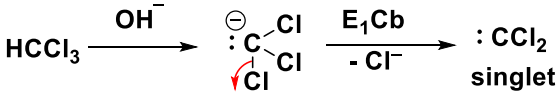


triplet due to conjugation

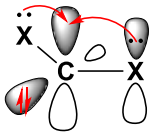


Singlet **M = 2S + 1**
= 2(1/2 - 1/2) + 1
= 1

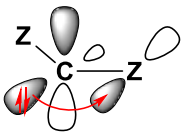
- **H C H bond angle 100 - 110°**
- **act as electrophile ⟶ ambiphile**
- **usually generated via ionic mechanism**



- **examples -$\ddot{C}Cl_2$, $\ddot{C}HCl$, $\ddot{C}(OM)_2$**



X = donor atom (eg. Cl, OMe) stabilizes singlet state by donating electron density to LUMO of carbene



X = acceptor (eg. CO_2Me) stabilizes singlet state by accepting electron density from HOMO

- Usually (for simple hydrocarbon) the singlet state is destabilized by ~ 8-10 kcal/mol than the triplet

⟹ pairing of electrons in singlet cost energy

- However, since the carbenes reacts very fast , they do not get the chance to rearrange to their most stable form.

- a reaction occurs by an ionic medium make singlet carbene.

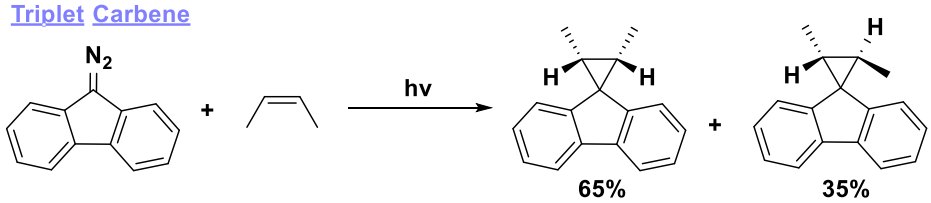
- in gas phase carbene is mostly triplet.

● **Singlet** —(**passing through inert gas**)→ **Triplet**

Triplet —(**Oxygen**)→ **Singlet**

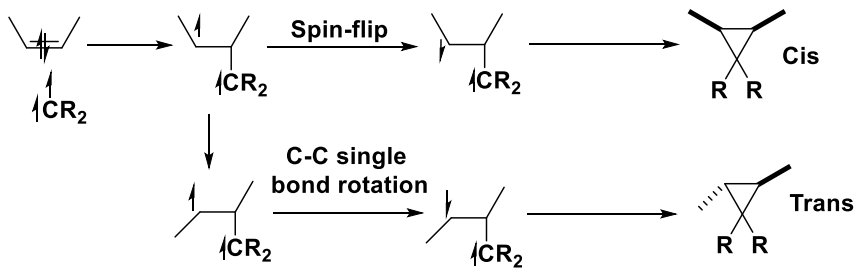
Reaction with alkenes :

Triplet Carbene



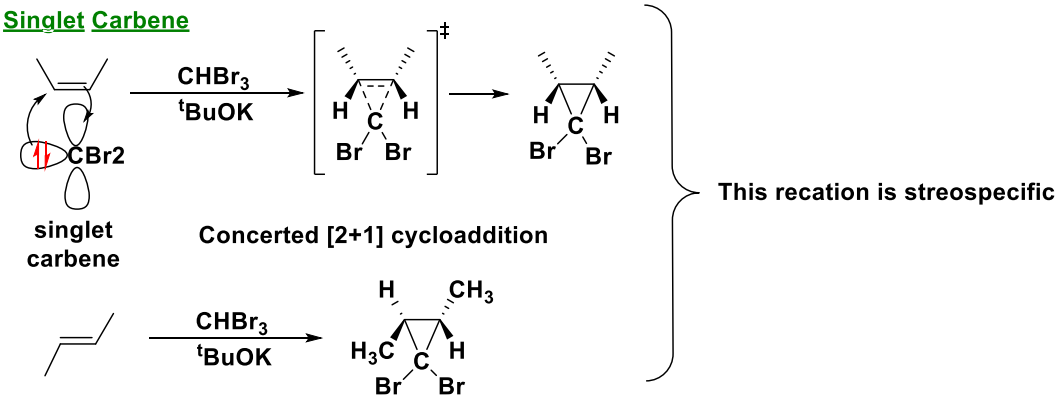
The reaction is not selective (not stereoselective, not stereospecific)

Mechanism:

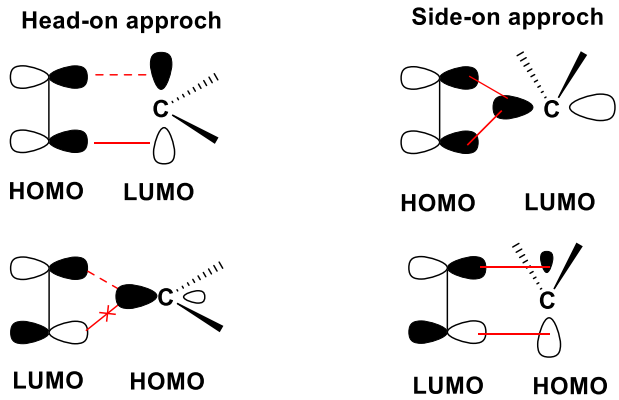


Here, the % of cis/trans will depends on the rates of spin-inversion vs C-C single bond rotation.

Singlet Carbene

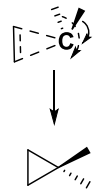


Orbital over view for the reaction of singlet carbene :



does not lead to
bond formation

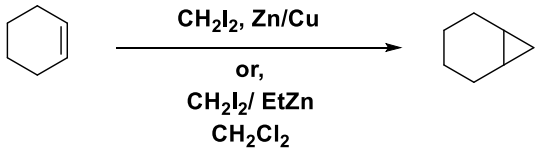
lead to
bond formation



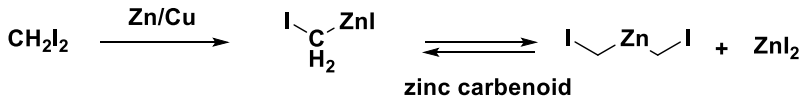
As the carbene approaches the TS it has to rotate.

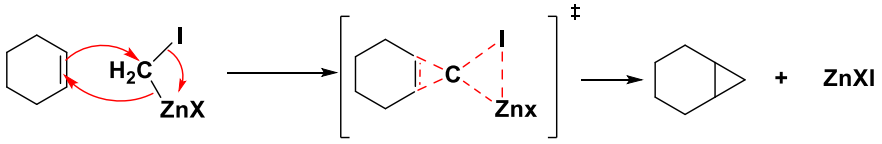
Class 6

Simon - Smith Reaction

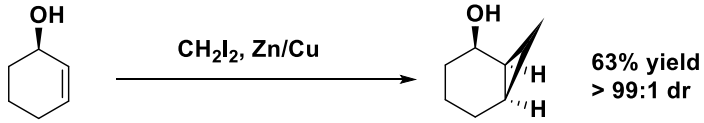


Mechanism :



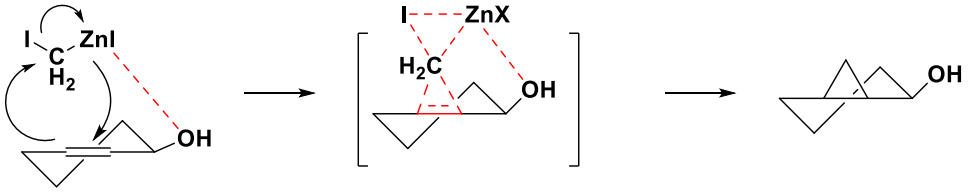


Diastereoselectivity:



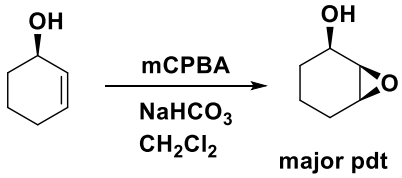
63% yield
> 99:1 dr

approach from the same face of the coordinating group

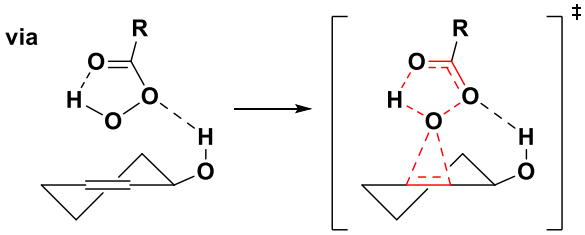


coordination at the same face of the OH

Similarly for the epoxidation reaction



major pdt
dr >99:1



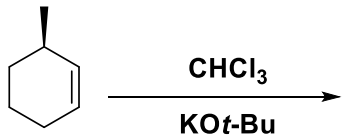
H-bonding at the same face of the OH

Class 6

QUESTIONS:

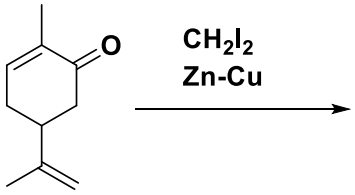
1. Draw the product(s) of the following reactions

a.

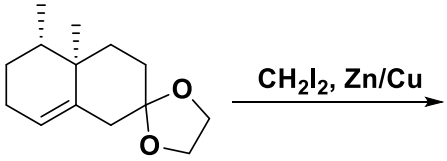


b.

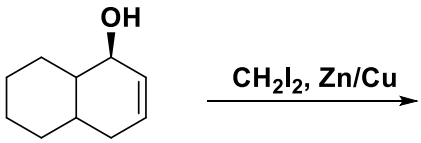
c.



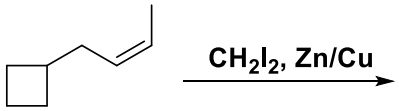
d.



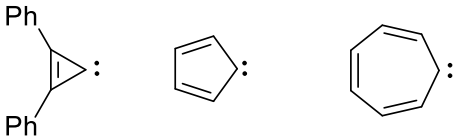
e.



f.



2. Carbene can be both electrophilic and nucleophilic. True or False? Justify.

3. Which are EPR active: (a) :CMe2 , (b) :CH2 , (c) :CCl2 , (d) :C(OMe)2, (e) :CF2

4. Predict the reactivity of the following carbenes towards diethyl fumarate and 1-hexene.